Bitcoin Dynamics: The *inverse square law* of market fluctuations and other stylized facts

Soumya Easwaran[1] and Sitabhra Sinha[2,3]

[1]The Institute of Mathematical Sciences, CIT Campus, Taramani, Chennai 600113, India
[2]The Institute of Mathematical Sciences, CIT Campus, Taramani, Chennai 600113, India
[3]National Institute of Advanced Studies, IISc Campus, Bangalore 560012, India.

E-mail: [1]soumyae@imsc.res.in, [2]sitabhra@imsc.res.in

Abstract

Analysis of time-series data of different markets have produced evidence for several *stylized facts* (universal features) including heavy tails characterized by power law exponents, which provide us tantalizing hints of the dynamics underlying such complex systems. It is especially important to see how these features evolve over time after the market is created and gradually develops. The recent advent of the digital currency, Bitcoin, and its growing popularity as an asset traded between agents over the last few years, provides us with an invaluable dataset for such a study. Bitcoin is a decentralized market and its value is not controlled by a single institution, (e.g. a central bank). Here we have analyzed high-frequency Bitcoin trading data (with a resolution of 1 tick, i.e., a single trading event). We show that the distribution of price fluctuation (measured in terms of logarithmic return) has a heavy tail. The exponent of the tail implies that Bitcoin fluctuations follow an *inverse square law*, in contrast to the *inverse cubic law* exhibited by most financial and commodities markets. The distribution of transaction sizes and trading volume are seen to have Levy-stable distribution. Multi-scale analysis show the presence of long term memory effects in market behavior.

Keyword: Bitcoin, Stylized facts, Heavy Tail, Power law, Inverse square law

Bitcoin is a decentralized software-based cryptocurrency developed by Satoshi Nakamoto (pseudonym) in 2008, and first used on January 3, 2009. It is a digital currency whose implementation is based on cryptography which is used to both validate transactions and generate new currency. Like most cryptocurrencies, the creation and transfer of bitcoins is based on an open source protocol which is independent of any central authority. In certain respects, Bitcoin is like cash in that both parties in a transaction are qausi-anonymous, known only by their pseudonyms. The parties are identified by an address, a 27-34 alphanumeric character that comprises a pair of public and private keys. Addresses can be generated by any user at no cost and it is thus possible for a user to have more than one address. The ownership of the existing bitcoins is identified with the public key and outgoing payments are signed by an owner using the private key.

Bitcoin network operates on a peer to peer model without any regulatory oversight. All transactions in the bitcoin ecosystem are recorded in a public ledger which is called the blockchain. Thus although all transactions are between anonymous real-world entities, every transaction is globally visible. As the blockchain is essentially the entire transaction history, each new payment is validated by checking its consistency with the blockchain and once it is verified by majority of peers in the network, is added to the blockchain. Anyone with an internet connection can download a client and become a part (node) of the network. This is achieved by having a computationally intensive process for verifying transaction history so that more nodes in the network imply a more secure network. The data analyzed here was obtained from the MtGox exchange.

The price of Bitcoins, P_t have shown a remarkable rise over the few years that it has been in existence and the price fluctuations as measured in terms of logarithmic returns $r_{t,\Delta t} = log(P_{t+\Delta t}/P_t)$ normalized by subtracting the mean value and dividing by the standard deviations shows large fluctuations over the entire period.
The distribution of r_t as for most financial markets, show a large deviation from the Gaussian distribution and is characterized by a heavy tail that follows a power law $P_c(r_{t,\Delta t}) \sim r_{t,\Delta t}^{-\alpha}$. However, in the case of Bitcoin, the power-law exponent α is quite distinct from 3 that is seen for most markets (this is the well-known *inverse cu-*

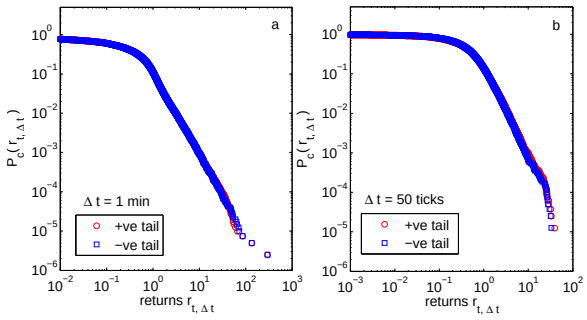


Figure1: Cumulative distribution of normalized returns for Bitcoin during Sept 2010-Jan 2014 calculated for (a) real time with time interval $\Delta t = 1$ min and (b) 'tick time' with $\Delta t = 50$ ticks. The tail exhibits power law scaling with an exponent whose maximum likelihood estimate is $\simeq 2$.

bic law). This *inverse square law* for fluctuations is marginally outside the Levy-stable regime, so that if returns are calculated over increasing values of Δt the Gaussian distribution can be recovered at large Δt.

If, instead of real time, we calculate the returns over intervals defined in 'tick time', i.e., in terms of a fixed number of transactions, we again observe the power law with Hill estimated exponents of both positive and negative tails being around 2. Thus the *inverse square* law of price fluctuations for the Bitcoin market seem to be remarkably robust with respect to the interval definition being used for measuring returns.

We observe that the trading volume distribution (i.e., the distribution of the total number of Bitcoins traded in a certain time interval) to be also characterized by a heavy tail. However, the distribution exponent is in the Levy-stable regime. We also note that the distribution of the transaction size, i.e., the number of Bitcoins traded in each transaction, happen to also have a Levy-stable distribution. One can expect that the trading volume distribution will have the same nature as the transaction size distribution, and in particular, that they have the same exponent if (i) the latter is Levy stable and (ii) successive transaction sizes are independent of each other.

While autocorrelation for returns decay quickly to noise level as expected for efficient markets, that for volatility (measured in terms of absolute return) shows extremely slow logarithmic decay with time, indicating the existence of *volatility clustering*.

Multifractal scaling in the Bitcoin market is indicated by anomalous scaling of higher moments of price changes with time: $M_q(\tau) = \langle |r_{\tau(t)} - m_\tau|^q \rangle$, where m_τ is the mean return calculated at scale τ.

It is remarkable that the return distribution exponent not only departs significantly from the "inverse cubic law" reported for returns of many financial and commodities markets - and widely believed to be universal - but it is also pointing the the occurrence of an "inverse square law" of markets, predicted by the model proposed in Ref. [1]. In this model, where N agents buy and sell an asset depending on the deviation of its instantaneous price from the long-term average (used as a surrogate for an estimate of the "fundamental" price of the asset), the return distribution exhibits a distribution with a heavy tail over a large range values of the parameter μ controlling the sensitivity of an agent to the magnitude of deviation of the price from its perceived fundamental value. It is of interest to note here that when agents are identical in terms of sensitivity to the price variation (i.e., every agent has the same μ (> 50, say) , the model predicts the inverse square law, whereas if the agents are heterogeneous (i.e., μ is distributed over an interval, say, [10,200]), the usual inverse cubic law is obtained. Comparison between the model results and the Bitcoin market suggests that the latter may be distinct from conventional financial markets in that the agents trading in Bitcoin may be relatively more homogeneous in terms of their risk-sensitivity. Given that in the initial stages of the Bitcoin market, only a relatively small section of population, viz., technophiles, traded in it, the assumption of homogeneous agents is perhaps not unreasonable.

References

[1] S. V. Vikram and S. Sinha, Emergence of universal scaling in financial markets from mean-field dynamics. *Phys. Rev. E* **83** 016101 (2011).